

Exploring the Spatial Distribution of Persistent SARS-CoV-2 Mutations - Leveraging mobility data for targeted sampling

Reviewed Preprint

Published from the original preprint after peer review and assessment by eLife.

About eLife's process

Reviewed preprint version 1

April 26, 2024 (this version)

Posted to preprint server

April 15, 2024

Sent for peer review

January 10, 2024

Riccardo Spott ✉, **Mathias W. Pletz**, **Carolin Fleischmann-Struzek**, **Aurelia Kimmig**, **Christiane Hadlich**, **Mathias Hauert**, **Mara Lohde**, **Mateusz Jundzill**, **Mike Marquet**, **Petra Dickmann**, **Ruben Schüchner**, **Martin Hölzer**, **Denise Kühnert**, **Christian Brandt**

Institute for Infectious Diseases and Infection Control, Jena University Hospital, Jena, Germany • Center for Sepsis Control and Care, Jena University Hospital/Friedrich Schiller University Jena, Jena, Germany • SMA Development GmbH - epicinsights Agentur für Künstliche Intelligenz und Big Data Analytics, Jena, Germany • Department of Anaesthesiology and Intensive Care, Jena University Hospital, Jena, Germany • Thuringian State Authority for Consumer Protection, Department Health Protection • Methodology and Research Infrastructure, Genome Competence Center (MF1), Robert Koch Institute, Berlin, Germany • Centre for Artificial Intelligence in Public Health Research, Robert Koch Institute, Berlin, Germany • Transmission, Infection, Diversification and Evolution Group, Max Planck Institute for Geanthropology, 07745 Jena, Germany

 https://en.wikipedia.org/wiki/Open_access

 Copyright information

Abstract

Given the rapid cross-country spread of SARS-CoV-2 and the resulting difficulty in tracking lineage spread, we investigated the potential of combining mobile service data and fine-granular metadata (such as postal codes and genomic data) to advance integrated genomic surveillance of the pandemic in the federal state of Thuringia, Germany. We sequenced over 6,500 SARS-CoV-2 Alpha genomes (B.1.1.7) across seven months within Thuringia while collecting patients' isolation dates and postal codes. Our dataset is complemented by over 66,000 publicly available German Alpha genomes and mobile service data for Thuringia. We identified the existence and spread of nine persistent mutation variants within the Alpha lineage, seven of which formed separate phylogenetic clusters with different spreading patterns in Thuringia. The remaining two are sub-clusters. Mobile service data can indicate these clusters' spread and highlight a potential sampling bias, especially of low-prevalence variants. Thereby, mobile service data can be used either retrospectively to assess surveillance coverage and efficiency from already collected data or to actively guide part of a surveillance sampling process to districts where these variants are expected to emerge. The latter concept proved successful as we introduced a mobility-guided sampling strategy for the surveillance of Omicron sublineage BQ.1.1. The combination of mobile service data and SARS-CoV-2 surveillance by genome sequencing is a valuable tool for more targeted and responsive surveillance.

eLife assessment

The authors analyze the relationship between human mobility and genomic data of SARS-CoV-2 using mobile phone mobility data and sequence data and present a **solid** proof of concept. This **useful** work was conducted on a fine spatial scale and provides suggestions on how mobility-derived surveillance could be conducted, although these results are mixed. The primary significance of this work is the strong use of large datasets that were highly granular. The authors provide a rigorous study, but with less clear predictive power of mobility to inform transmission patterns.

Introduction

On March 11, 2020, the World Health Organization (WHO) classified the SARS-CoV-2 virus (Severe Acute Respiratory Syndrome - Corona Virus - 2) as a global pandemic due to its rapid spread and high infection rate.¹ The airborne virus has since caused significant morbidity and mortality worldwide (<https://covid19.who.int/>). In an attempt to control its spread, many countries initiated comprehensive surveillance efforts with molecular techniques such as Polymerase Chain Reaction (PCR) and Whole Genome Sequencing (WGS).^{2,3} Consequently, nearly 15.8 million SARS-CoV-2 sequences have been deposited into the “Global Initiative on Sharing All Influenza Data” database (as of July 21, 2023, GISAID). Many research groups have undertaken studies examining the viral spread by integrating sequencing and epidemiological data to monitor the pandemic and investigate local outbreaks.^{4,5} Most of these local projects are part of national surveillance programs such as the UK’s Genomics Consortium (COG-UK) or “national genomic surveillance” in the USA.^{2,6} In Germany, the “Coronavirus-Surveillanceverordnung” (CorSurV) enacted by the State Ministry of Health on January 19, 2021, mandated that laboratories with sequencing capabilities process SARS-CoV-2-positive samples, offering financial compensation until April 2023.³

Bioinformatics workflows developed in Germany, such as poreCov (for Oxford Nanopore data) and CoVpipe2 (for Illumina data), reconstruct SARS-CoV-2 consensus genomes from the sequencing data and prepare the results for upload and submission to the Robert Koch Institute (RKI).^{7,8} As the German government’s public health and biomedical research institute responsible for disease control and prevention, the RKI collected the genomes via the German Electronic Sequence Data Hub (DESH) and integrated them with additional epidemiological information to provide an up-to-date overview of the ongoing viral spread. For keeping track of the rapid SARS-CoV-2 evolution, PANGO (Phylogenetic Assignment of Named Global Outbreak) provides a standard naming convention based on unique mutation profiles and further criteria, resulting in the classification of over 3,660 lineages (as of August 2023).^{9,10} Additionally, the WHO classified important viral lineages as “Variants of Concern” (VOC), “Variants of Interest” (VOI), or “Variants under Monitoring” (VUM), using Greek designations in the past (e.g., “Alpha” (Pango lineage main designation B.1.1.7) or “Omicron” (Pango lineage main designation B.1.1.529)). Further, the WHO also de-escalated former VOCs to reflect the current SARS-CoV-2 variant landscape better. The first defined VOC (now de-escalated), the Alpha lineage, rapidly replaced almost all previously circulating lineages globally by the end of 2020 until the VOC Delta (main lineage B.1.617.2) replaced it in mid-2021.^{11–13}

To predict or monitor the rapid viral spread throughout regions, various data types, like travel data, passenger volumes, or passive wastewater monitoring were examined previously.^{14–16} Furthermore, different studies explored mobility data with genomic data to retrace the origin and spatial expanse of Alpha or utilized geolocation data to model the spread in metropolitan areas to

recreate case trajectories and the impact of mobility restrictions.^{17,18} Mobility data was also used in Germany during the pandemic, revealing that lockdowns leave distant parts of the country less connected due to the sharp decline in long-distance travel.¹⁹ These studies focused on analyzing residential movement and contact tracing to evaluate and inform health policies but were not applied to active molecular surveillance.

Here, we investigated whether mobile service data and fine-granular metadata (such as postal codes and genomic data) can help predict the spread of the Alpha lineage or guide the sampling for more targeted genomic surveillance with a focus on the German federal state of Thuringia.

Results

The Alpha lineage spread rapidly through Thuringia, showing a pattern similar to its nationwide spread

Thuringia is a rural federal state in central Germany with a population of 2.1 million and no major airports (overview of Thuringia's population density in Supplementary Figure S1). We investigated if the spread of the Alpha lineage of SARS-CoV-2 behaved differently compared to the whole of Germany. To understand its spread, we used 257,721 public SARS-CoV-2 genomes from Germany (excluding Thuringia; including 136,099 Alpha genomes) and 7,404 genomes from our own sequencing data for Thuringia (including 6,522 Alpha genomes) from December 2020 to August 2021 (see **Figure 1**, Supplementary Figure S2, and Supplementary Tables S1 and S2; for details, see Methods section “Alpha spread datasets”). For Thuringia, district-level data (full postal code) per genome were available, whereas, for Germany, only postal code data of the sending laboratories (referred to as “primary diagnostic laboratory” by the RKI where the SARS-CoV-2 positive sample was detected) and sequencing laboratories were publicly available.

In late December 2020, four federal states in Germany (from here on called states) reported the first cases of the Alpha variant. Although sequencing was initially low, it gradually increased in the following month. However, the Corona-Surveillance regulation was passed at the end of January 2021, leading to a rapid increase in sampling and sequencing by February since sequencing costs could be reimbursed. Even though Thuringia sequenced a similar amount of SARS-CoV-2 samples compared to other German states (as shown in **Figure 1**), the proportion of the Alpha variant to other lineages was relatively low. However, the proportion of Alpha increased heavily in February.

By March, Alpha had spread to nearly all states and districts (districts are similar to counties or provinces) in Germany (Median: 76.14 % compared to 35.92 % in February, excluding Thuringia) and Thuringia (Median: 85.29 %, up from 50.00 % in February). So, there was no noticeable difference in the Alpha proportions between Germany and Thuringia after February. During the summer of June and July 2021, sequencing declined in Germany (including Thuringia; Supplementary Figure S2) due to the decrease in overall daily cases, as reported by Meintrup *et al.* and Oh and Hölzer *et al.*^{20,21}

In summary, the spread of the Alpha lineage in Thuringia lagged roughly two weeks behind the general spread in the rest of Germany but showed similar proportions. This suggests that Thuringia experienced a delay in the initial arrival of Alpha. However, we did not observe any difference in the overall spread afterward. Thuringia was among the first states to adopt new containment measures, including contact limitations, closure of retail shops, and prohibition of tourist journeys (14th December 2020). Jena, a city in Thuringia, was also the first German city to implement mandatory public masking in March 2020.²² Contacts were further restricted on January 9th, and people were urged to restrict their movement radius to 15 km, which might explain the delay besides the absence of major airports nearby.

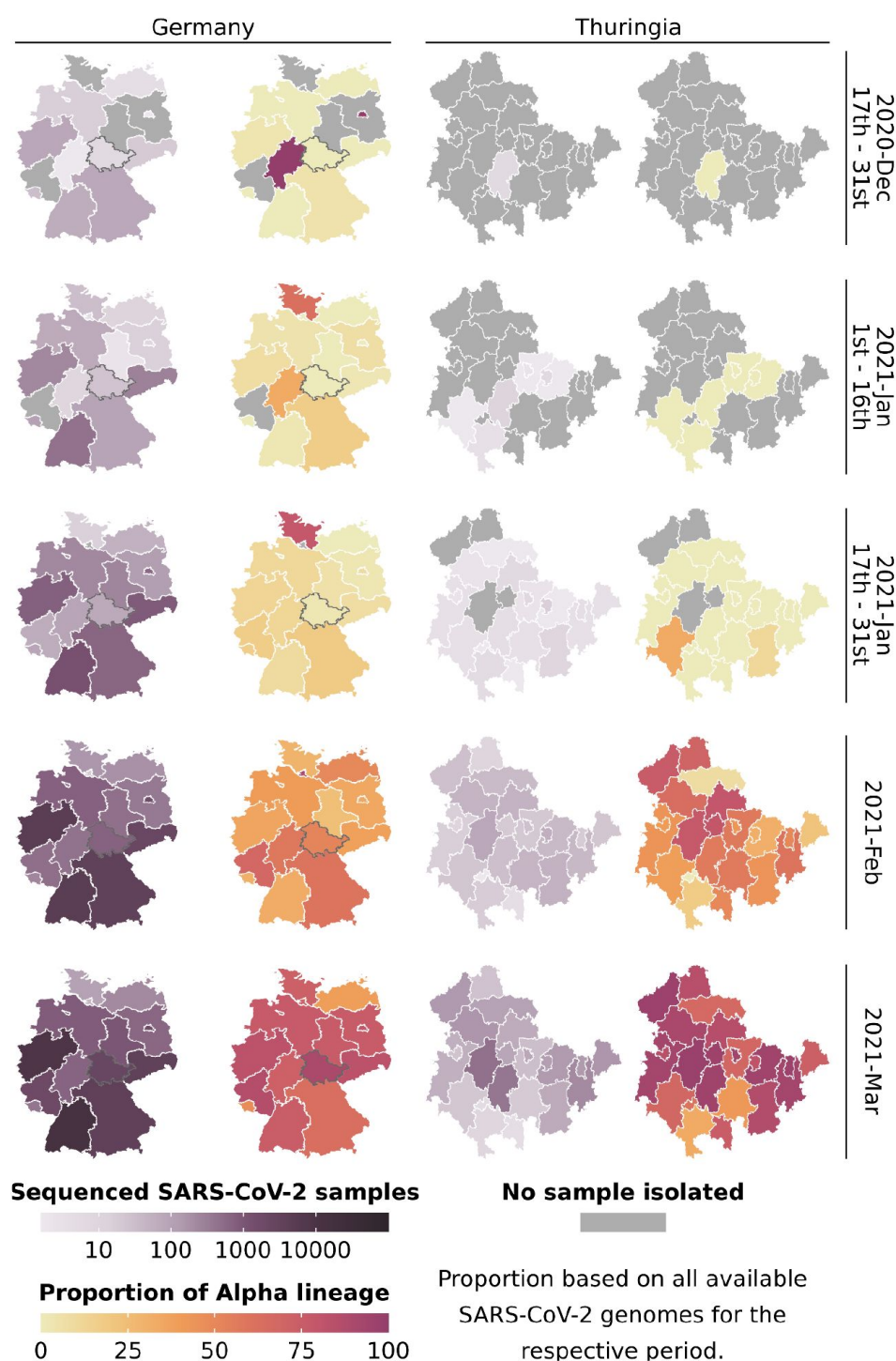


Figure 1

Total number of all sequenced SARS-CoV-2 samples (purple) and the proportion of the Alpha lineage for all sequenced samples (yellow-red) for each state of Germany and each district of Thuringia.

257,721 publicly available German SARS-CoV-2 genomes and their metadata were used for the general German maps excluding data from Thuringia. For Thuringia, we always used 7,404 genomes and their metadata from our database for both the German and the Thuringian maps. Please note that for all states except Thuringia, we used the postal code of the sending laboratory as a proxy for the geographic location of a sample. Thuringia is highlighted by a grey border on the maps of Germany.

All Thuringian genomes were evenly distributed between other German samples in the phylogenetic time tree (see Supplementary Figure S3). However, due to its rapid spread from February onwards, it is difficult to accurately track how the Alpha lineage specifically expanded (point of entries, exact origins, etc.). Consequently, we investigated whether “sublineages” might be identifiable and trackable to address this.

Monitoring of Alpha subclusters in Thuringia reveals temporally and regionally restricted distribution patterns

To identify possible clusters among the Alpha lineage spreading in Thuringia, we called each Alpha genome’s mutations via Nextclade by analyzing them using poreCov^{7,23}. We identified nine clusters out of 70,429 Alpha genomes, based on their mutation profile, time period, and phylogenetic distance (from here on called Alpha subclusters; for details, see Methods “Subcluster identification”). All subclusters, their time period, and sample size in Thuringia are summarized in **Table 1**. An overview of each subcluster (phylogenetic time tree, location, and period) is also provided on *microreact.org* as interactive views (see Methods “Subcluster identification”). Note that our subcluster definition is similar to the definition of a sublineage. However, during the Alpha wave, PANGO sublineages were only rarely defined (PANGO designation: Q.1 to Q.8; compared to the Delta and Omicron waves).

Eight of these subclusters are based around a specific spike protein mutation, while the other contains a mutation within the ORF1b region. The subcluster 7.1 “S:N185D” branched out from the subcluster 7 “ORF1b:A520V” and subcluster 6.1 “S:V90F” is a branch of subcluster 6 “S:S939F” (see **Table 1**). These two branched subclusters still carry the specific mutation of their originating subcluster. The subclusters 3, 4, and 5 were observable between two and three months, and the other subclusters over at least four months. To investigate these subclusters’ regional spread, each sample was mapped to its Thuringian district based on the resident’s postal code it was isolated from. We then sorted the samples according to their subclusters and visualized them throughout the subcluster’s observed period. The spread of two representative subclusters is exemplary visualized in **Figure 2a**, and all the subclusters are available via Supplementary Figure S4 and their data in Supplementary Table S3. Additionally, all subclusters and their metadata are also available via Microreact (see Methods “Subcluster identification”).

Each of the seven main mutation variant clusters originated from a different Thuringian district. At the same time, two subclusters, 6.1 and 7.1, branched out from the same districts as their original clusters (6 and 7), 12 or 13 days after their first emergence. The subclusters mainly spread regionally confined and not across all of Thuringia (see **Figure 2a**) but were also identified in other states of Germany (see “microreact.org”-project). For example, the “S:S939F” subcluster spread across 15 states, with the first samples being isolated outside of Thuringia. The eight Spike-mutation subclusters had expanded between four to twelve of the 23 Thuringian districts within the observable time period of each subcluster. They expanded by one to six districts per month, with a greater expansion mostly accompanied by a larger increase in the subcluster sample number. In contrast, the ORF1b-variant even comprised 21 districts and expanded between two to seven districts per month. Most of each subcluster’s samples were identified in their region of first occurrence, and no additional samples were found after the given periods.

Several limitations need to be considered. The identified subclusters may have multiple origins or may not originate from Thuringia. Due to the lack of precise zip codes (publicly available German genomes are limited to postal codes of sending and sequencing laboratories), monitoring the subclusters in other states on a district level was impossible. Nevertheless, we could follow how the subclusters developed in Thuringia, even if multiple origins may have affected the overall speed or length of each subcluster’s occurrence.

Designation	Mutations	Number of samples	Time period	Remarks
1	S:H49Y , ORF1a:I841V	44	Feb-May 2021	S:H49Y eases cell entry in S-pseudotyped lentiviral system. ²⁴
2	S:N354K	63	Feb-May 2021	S:N354K slightly impaired mAb h11B11. ²⁵
3	S:G496S , ORF1a:E1013K	12	Mar-May 2021	S:G496S: compromises BA.1 replication fitness and changed mAb sensitivities, reduces ACE2 binding affinity, and increases immune evasion. ²⁶⁻²⁸
4	S:N703D , ORF1a:D1228G, ORF1a:A2123V	51	Mar-May 2021	-
5	S:T716V , N:G204P, ORF1a:D1600N	22	Apr-May 2021	-
6	S:S939F	206	Feb-May 2021	S:S939F: modulates T-cell propensity. ²⁹
6.1 [#]	S:V90F , S:S939F	55	Feb-May 2021	-
7	ORF1b:A520V	811	Feb-Jun 2021*	-
7.1 ^{\$}	S:N185D , ORF1b:A520V, ORF1b:L1504F	40	Feb-May 2021	-

* only one sample for June # branch from subcluster 6 \$ branch from subcluster 7

Table 1

Overview of nine Alpha subclusters in Thuringia, their sample count, their time period, and their specific mutations that are shared across all members of the subcluster (excluding characteristic Alpha mutations that are shared across all subclusters).

The mutation used to define the subcluster is highlighted in bold.

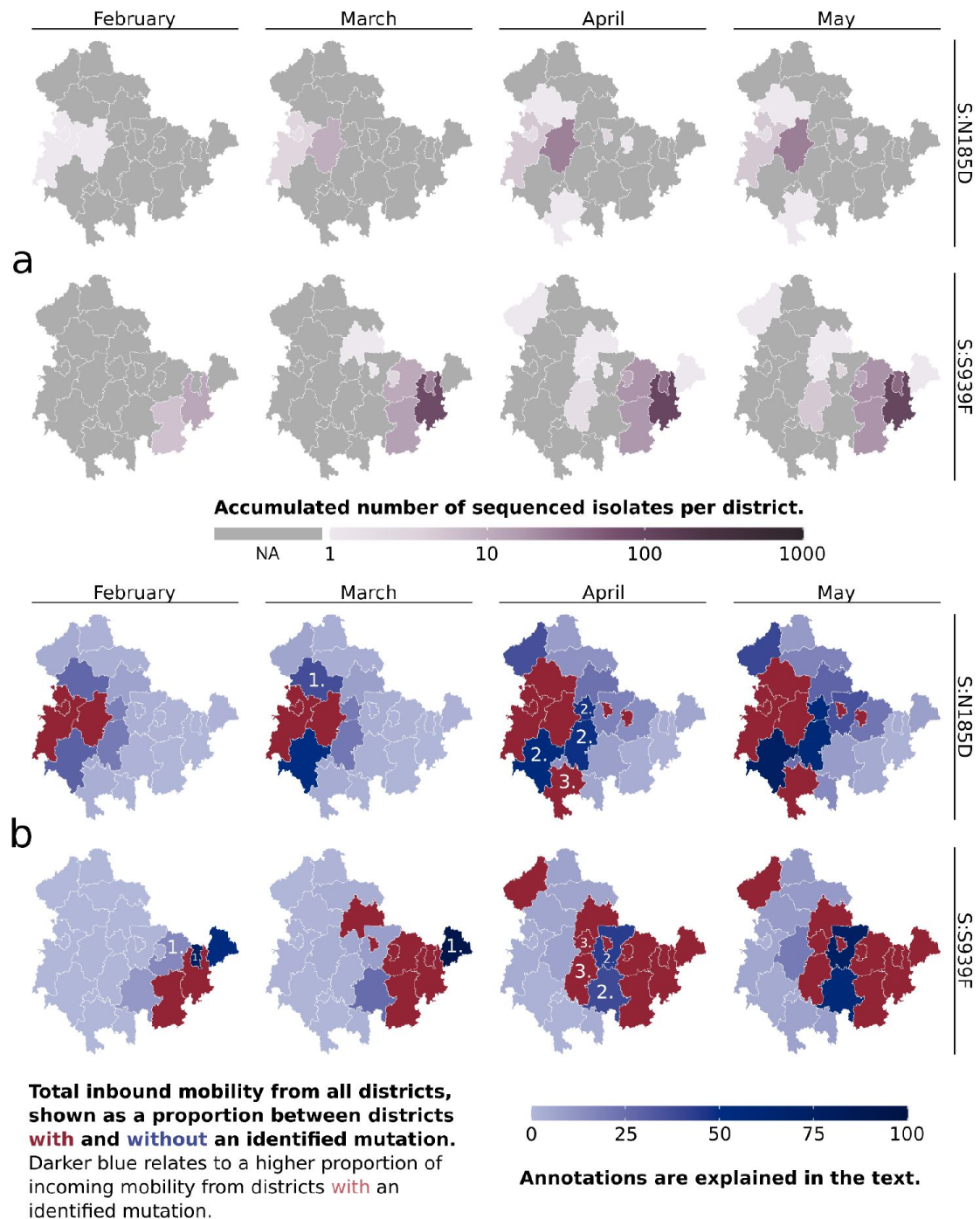


Figure 2

Overview of the subclusters "S:N185D" and "S:S939F" in Thuringian districts.

a) Accumulated number of sequenced samples for each subcluster per district and per month. b) Combined visualization of each district's "inbound mobility" from other districts (color intensity) and the occurrence of a subcluster sample (red = sample found, blue = no sample found). The inbound mobility of each district (color intensity) is shown as a proportion of incoming mobility from other districts with or without an identified sample. The darker the blue color of a district, the higher the proportion of inbound mobility from other districts with an identified subcluster sample (red districts). The light blue color describes that most of the inbound mobility of a district comes from other districts without an identified subcluster sample (blue districts). Numbers refer to district types 1, 2, and 3, as further defined in the main text.

Our surveillance sampling heavily relies on various institutions and partners, and only a portion of the provided samples can be sequenced (see “Sampling” in Methods). For example, the spread of subcluster “S:S939F” revealed two districts in April where no respective samples were found (**Figure 2a**) despite being surrounded by districts with “S:S939F”-samples present. This could be due to the lack of samples sent to sequencing from those regions or the low prevalence. We, therefore, investigated if mobile service data of residents, in addition to molecular surveillance, might be utilized to counteract this issue.

Mobile service data indicates Alpha subcluster spread and sampling bias

With the aim to predict the subcluster spread and, thereby, reduce surveillance-based sampling bias, we utilized anonymized mobile service data from T-Systems International GmbH. Around 200 million trips were used to determine the number of daily trips between the Thuringian districts. We then combined this information with our fine-granular genomic data to specify each district’s monthly proportion of inbound mobility from subcluster-affiliated districts (see Methods “Mobile service data”). The results are visualized in **Figure 2b** (complete overview in Supplementary Figure S5; data provided in Supplementary Table S4).

The addition of mobile service data resulted in three different “types” of districts (see **Figure 2b**, annotated districts). Type 1 included districts with high inbound mobility from areas with an identified variant, where the variant was eventually found afterward, while Type 2 were districts with high inbound mobility from areas with an identified variant, where the variant was never identified. Type 3 included districts not directly connected to a district with an identified variant, but a variant was eventually identified while they border Type 2 district(s). Our previous analysis of the subclusters’ spreading pattern across the districts, based solely on identified variants, indicated missed identifications in some districts due to the seemingly illogical spread to districts without a connection to others (**Figure 2a**). The inclusion of mobile service data revealed some of these districts to be Type 2 districts. This suggests that the specific variant should be identifiable within these districts due to the observed high incoming mobility from districts with identified variants. Type 2 districts were mainly observed for subclusters with low prevalence and, consequently, low numbers of covered samples that are usually more difficult to monitor. For example, we assumed missing identifications in some districts of subclusters 1, 2, and 3, which through the mobile service data, are now partially identified as Type 2 districts. In contrast, for fast-spreading, highly prevalent subclusters, the regional coverage aligned well with the mobile service data, such as subcluster 7 covering 811 samples (see Supplementary Figure S5). Despite analyzing the mobile service data of districts from other federal states than Thuringia, we could not apply them, as the lack of precise location data for samples outside of Thuringia prevented the correct calculation of the incoming mobility. Based on the nine observable clusters, we concluded that mobile service data might be a good prediction marker for the spread of high-prevalence variants but, more importantly, a good indication of districts that should have an identified low-prevalence variant. Next, we investigated if mobile service data can improve active surveillance via guiding sample collection for genomic sequencing.

Proof of principle: Mobile service data-guided sampling for genomic surveillance for Omicron BQ.1.1

Based on our previous findings, we implemented the “mobility-guided” sampling approach under real pandemic circumstances over one month in addition to our active surveillance.

As the subject of investigation, we searched for a newly emerging (based on global news reports) and ideally low prevalent SARS-CoV-2 lineage in Thuringia.

Among the various emerging Omicron sublineages during that time, sublineage BQ.1.1 fulfilled the defined criteria. First isolated in a northwestern-Thuringian community on October 5, 2022, we identified this particular sublineage on October 14, 2022, among a routine batch of 42 samples. BQ.1.1 was a low-prevalence sublineage that was identified worldwide (<https://outbreak.info/situation-reports?pango=BQ.1.1>). Following its first Thuringian identification, we utilized the past two years of mobile service data (2020/21) to investigate the residential movements for the community of first detection. As a result, we identified eight communities with the most residential movement from the originating community (four in central and three in NW of Thuringia, one in NW-neighboring state Saxony-Anhalt; purple arrows in **Figure 3**, “Guided sampling” in October 2022). Subsequently, we specifically requested all the available samples from these communities and collected 19 additional samples (isolated between the 17th and 25th of October 2022) besides the randomized sampling strategy. More samples could not be included as sampling was restricted to the submission of third parties we had no influence over. As part of the general Thuringian surveillance, we collected 132 samples for October (covering dates between the 5th and 31st) and 69 samples in November (covering dates between the 1st and 25th; see “Randomized Sampling” for the according months in **Figure 3**). Randomized sampling was not influenced or adjusted based on the mobility-guided sample collection. A complete overview of all samples is provided in Supplementary Table S5.

Among the 19 samples specifically collected based on mobile service data, we identified one additional sample of the specific Omicron sublineage BQ.1.1 close to the originating community. During the same period, our randomly sampled routine surveillance strategy did not detect another sample (October 2022, **Figure 3**). Only in the one-month follow-up, another four samples were identified across Thuringia through routine surveillance (November 2022, **Figure 3**). During our attempt to implement the mobility-guided sampling approach in real-time during the pandemic, we encountered three distinct limitations, some of which are commonly observed in surveillance practices. The guided sampling depended on the individual sample submitting institutions, affecting the availability of suitable samples, especially for the communities of interest. By choosing a newly emerging Omicron sublineage for our experiment, spread and, therefore, suitability were uncertain. In our case, BQ.1.1’s prevalence in Thuringia was even lower than expected, also remaining rare in subsequent months, with only 42 samples found until June 2023, eight months after the first occurrence in Thuringia. Due to the short preparation time, only mobile service data from the past two years and no current data were available. Nevertheless, the available datasets still reflect pandemic movement behavior since the pandemic was already ongoing for two years. In summary, increasing the sampling depth in the suspected regions resulted in successfully identifying the specified lineage using only a fraction of samples in contrast to the randomized surveillance. Implementing such an approach effectively under pandemic conditions poses difficult challenges due to the fluctuating sampling sizes. Although the finding of the sample may have been coincidental, our proof of concept demonstrated how we can leverage the potential of mobile service data for targeted surveillance sampling.

Conclusion

During the SARS-CoV-2 pandemic, diverse data sources like travel, wastewater, and mobility data have been employed in surveillance and transmission tracking.^{14–19} In the present study, we analyzed over 265,000 German SARS-CoV-2 genomes to examine whether mobile service data can predict the spatial distribution of the Alpha lineage in the German state of Thuringia and how they potentially benefit pandemic surveillance.

Our study shows that the absence of major transport hubs in Thuringia initially delayed the spread of Alpha. However, its impact on the total distribution is limited, and the spread was ultimately comparable between Germany and Thuringia. While our findings on mobile service data may, therefore, also apply to Germany, we could not verify this because the limited location

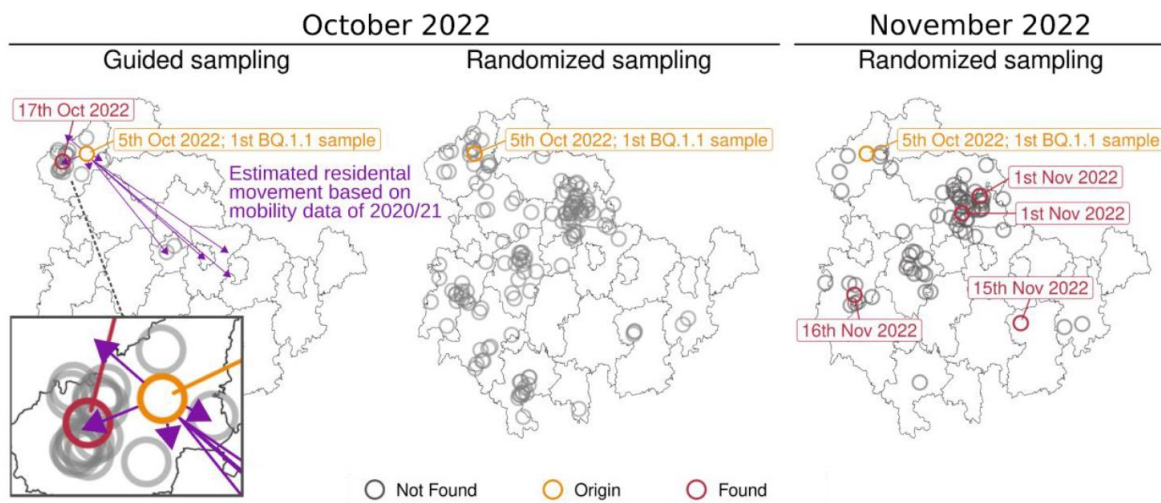


Figure 3

Overview of the mobility-guided sampling of the Omicron sublineage BQ.1.1 in Thuringia compared to the default randomized sampling (surveillance) in October 2022.

For clarity, the surveillance results in November 2022 were added to highlight the spreading progress of BQ.1.1. Circles reflect the location of each sample (based on residents' zip codes). Orange circle: First identified BQ.1.1 sample; Red circle: Additionally identified BQ.1.1 sample; Grey: Sample of another lineage. Purple arrows show the eight residential movements with the most participants from the originating community of the first identified BQ.1.1 sample (orange circle). The mobile service data were extracted from 2020/21.

data of publicly available German genomes prevented in-depth investigations outside of Thuringia. Thus, precise patient location data are crucial to utilize mobile service data in genomic surveillance, but privacy regulations may restrict access to this data. Shortly after its emergence, Alpha formed mutation variants like the known sublineages Q.1 to Q.8 and the Thuringian subclusters identified by us. This reflects the ongoing evolution during active circulation and indicates an even greater sublineage diversity, which has not been surveyed as closely as in the subsequent Delta and Omicron waves. By monitoring the nine Thuringian subclusters, rather than focusing solely on the parental lineage B.1.1.7, we were again able to effectively track transmissions and gain a comprehensive understanding of the regional spread. So, it underscores the importance of sequencing in pandemic surveillance to explore such genomic changes and, thereby, keep track of the transmission chains and potential outbreaks.

Mobile service data can support such surveillance in different ways. Previous studies examined the capabilities of mobility data in the context of, e.g., case trajectories, but retrospectively applied to already collected data, it can be used to examine surveillance sampling coverage and possible sampling bias. We highlighted this approach exemplary with the Alpha lineage, where mobile service data indicated an assumed sampling bias and partially predicted the spread of our Thuringian subclusters. Another approach is to actively guide the sampling process through the usage of mobile service data, which we demonstrated with our proof of principle focusing on the Omicron-lineage BQ.1.1. The allocation of surveillance resources for guided sampling can be flexibly adjusted to adapt to specific circumstances and maximize efficiency. This allows for an increased regional sampling coverage without increasing the general sampling volume, which especially helps identify low-prevalence variants. We recommend using mobile service data as a supportive element for general randomized surveillance to retrospectively evaluate its efficiency and actively help screen for low-prevalence variants. These findings should also apply to other surveillance efforts. Yet the feasibility depends on the availability and cost of such mobile service data. Alternatively, financial resources could also be invested directly in increasing sampling capacity and coverage, which ultimately depends on individual factors of the respective surveillance. Mobile service data can also be used with other surveillance approaches and elements. For example, wastewater surveillance can give further indications to supplement guided sampling. At the same time, passenger data offers additional insights into traffic hubs as sources of regional movement.

Methods

Sampling

Starting mid-2020, we initially sequenced hospital-intern samples, transitioning by January 2021 to approximately 43 PCR-positive samples per week: 20 from the hospital's microbiology department and 23 randomly sourced by the Thuringian State Authority for Consumer Protection ("Thüringer Landesamt für Verbraucherschutz"; TLV).

Until June 2023, our institute sequenced 3,770 SARS-CoV-2 samples, and SYNLAB Holding Deutschland GmbH, Bioscientia Healthcare GmbH, and DIANOVIS GmbH provided additional 7,800 Thuringian SARS-CoV-2 genomes and their metadata.

Sample preparation and sequencing

RNA isolation used the ZymoResearch "Quick-RNA Viral Kit" (Zymo Research Europe GmbH, Germany, Product-ID: R1035), according to the manufacturer's instructions with 100 µl patient sample input and a centrifuge speed of 16,000 g.

The viral RNA underwent a Reverse Transcriptase (RT)-PCR followed by a multiplex-PCR using the ARTIC V1200 primer set, according to Freed and Silander's SARS-CoV-2 sequencing protocol (version 4, updating to version 5 by March 2021).³⁰ Subsequent DNA quantification utilized the Qubit dsDNA HS assay (Invitrogen, USA).

From the amplified DNA, a sequencing library was prepared using the Nanopore SQK-LSK109 and SQK-RBK004 kits (Oxford Nanopore Technologies, Oxford, UK), sequenced for a maximum of 72 h utilizing an Oxford Nanopore MinION Mk1b sequencer with R.9-flowcells and the MinKNOW software (versions MKE_1013_v1_revBC_11Apr2016 to MKE_1013_v1_revBR_11Apr2016 in the respective period), and analyzed with the software pipeline poreCov (versions 0.3.5 to 0.11.7; including basecalling, demultiplexing, adapter removal, quality filtering, and genome alignment) to reconstruct consensus genomes.⁷ Sequencing data and the respective metadata (e.g., isolation date, sending laboratory details) were submitted to the RKI through DESH. We also collected the postal code of the isolation location or at least of the sending local health authority, storing all data additionally in a local database.³¹ Due to data protection, such data is limited on the RKI's public GitHub repository (https://github.com/robert-koch-institut/SARS-CoV-2-Sequenzdaten_aus_Deutschland), providing instead postal codes of the sequencing and sending laboratories.

Alpha spread datasets

From our local database, we extracted 8,397 samples with isolation dates before Oct 1st, 2021. After adding federal state and district information, 993 entries with non-Thuringian locations were excluded, yielding 7,404 samples (including 6,522 Alpha genomes).

The publicly available RKI SARS-CoV-2 dataset was downloaded, containing 1,091,655 genomes with the respective metadata (17th Oct 2022; Zenodo-version 2022-10-16).³² 789,405 entries, isolated after Sep'21, and 64 entries without "sending laboratory" information were removed. For the resulting 302,186 entries, location information (location, federal state, district, longitude, latitude) were added based on the sending laboratory postal code. Five entries with a non-existing postal code and all 44,465 Thuringian samples were removed from the dataset, resulting in 257,721 samples (including 136,099 Alpha genomes). Analyzing both datasets, we calculated the monthly proportion of Alpha lineage samples in Thuringia and Germany per state/district, dividing Dec'20 and Jan'21 into first and second halves.

Subcluster identification

Using a total of 70,429 German and Thuringian Alpha genomes, a phylogenetic time tree was created (see Supplementary Method "Phylogenetic time tree construction" and Supplementary Figure S3). We determined the frequency of all non-Alpha-specific mutations among the 6,522 Thuringian Alpha genomes. We then screened for mutations present in at least 20 genomes with a small phylogenetic distance and a time occurrence of at least two months. This led to nine mutations, each of them creating a defined cluster covering between 12 and 811 closely related genomes. We only kept mutation information of these nine subclusters in the respective metadata, which, together with the tree file of the phylogenetic time tree, was uploaded to a "microreact.org"-project, provided as Supplementary File 1 and found under the following link: <https://microreact.org/project/ftR2GjfF6iXtSwbmN4ARTx-thuringianalpha-linclusters#76ir-complete-overview>.

Mobile service data

T-Systems International GmbH collected and aggregated mobile service data via the Cell ID method, dividing a geographical area into so-called traffic cells. Each cell is assigned to exactly one transmitter mast, with a spatial resolution from 500 m x 500 m up to 8 km x 8 km (depending on the transmitter mast network density). Cell phones always register to the closest traffic cell, which is recorded and stored in an Origin-Destination Matrix (ODM). For population representation, the

data was extrapolated with Deutsche Telekom's market share. Due to data privacy, the registration data is combined into movement streams between traffic cells, the status resolution is reduced to one hour (greater time intervals = less resolution), and individual traffic cells are grouped into districts. The degree of anonymization (k-value = 30, data privacy regulation) removed movement streams with less than 30 participants, resulting in approximately 200 Mio trips in the ODM. SMA Development GmbH analyzed all movements between the single Thuringian districts, adding each Alpha sample's isolation time and location data (per subcluster). The movements were further divided by months and originating district (subcluster-affiliated vs. -unaffiliated), determining each district's monthly inbound mobility proportion from cluster-affiliated districts.

Research in Context

Evidence before this study

We searched Pubmed for studies about the use of mobile service data for surveillance written in English. For the broadest possible search, we included any publication covering mobile data and surveillance aspects, using the following search string: ("cellular data" OR "cell phone data" OR "mobility data" OR "movement data" OR "migration data" OR "phone data") AND ("Surveillance" OR "Monitoring" OR "Survey" OR "Pandemic" OR "Disease" OR "Epidemic" OR "Outbreak"). Our search yielded 1,285 publications published between 1966 and 2023. We manually screened all these publications but found no study that applied mobile service data for active, targeted surveillance. Across all studies, the general focus was on tracking contacts or analyzing movements to assess, for instance, the efficiency of non-pharmaceutical interventions or generate prediction models. Some studies suggested targeted surveillance based on their results, but it was not yet applied. Additionally, we used "suite.ai" and "chatGPT" (with BING-search access) to let them search for "studies that utilize mobile service data to guide the sampling process for infectious disease surveillance". While "suite.ai" found two studies and "chatGPT" found another ten studies and reviews, none covered the direct application of the mobility data in active surveillance.

Added value of this study

This study highlights the value of combining mobile service data with fine-granular metadata for integrated genomic surveillance during the SARS-CoV-2 pandemic in a German federal state. We illustrated this strategy with the Omicron sublineage BQ.1.1 and how to guide the sampling processes toward areas where the new variant was expected to emerge. Additionally, we used mobile service data during the pandemic to assess our sampling coverage. Our study is the first to actively guide part of the genomic surveillance process during a pandemic.

Implications of all the available evidence

Efficient molecular surveillance setups are crucial in managing outbreaks from the local to the global scale. Different data sources are investigated to increase this efficiency, addressing factors like the more efficient usage of scarce surveillance resources and the prediction of spread. Extending molecular surveillance with such data should improve the future management of pandemics and outbreaks.

Data Availability

All genomic data (genomes and respective metadata) are available in the provided microreact project (project file available under <https://osf.io/n5qj6/>; <https://microreact.org/project/ftR2GjfF6iXtSwbmN4ARTx-thuringianalpha-linclusters#76ir-complete-overview>). The mobile service

data used in this study can only be published in processed form (available under <https://osf.io/n5qj6/> ). The original mobile service data can not be made public due to legal reasons/ownership. As the collector of the original aggregated mobility data, T-Systems applies to the EU- and German Data protection regulations.

<https://osf.io/n5qj6/> 

Author contributions

Sample collection and preparation, R.Sp, R.Sc., M.L., and M.M.; sequencing, R.Sp., M.L., and M.M.; database setup and maintenance, M.J.; software, C.B.; bioinformatic analysis, R.Sp., M.L., M.M., and C.B.; mobile service data analysis, C.H., and M.Ha.; literature research, R.Sp.; writing first draft, R.Sp.; reviewing and editing manuscript, R.Sp., C.B., M.M., C.F.-S., A.K., C.H., M.Ha., M.Hö., D.K., R.Sc., P.D., and M.W.P.; supervision, C.B.; project administration, C.B.; funding acquisition, M.W.P. All authors have read and agreed to the published version of the manuscript.

Funding

This work was supported by grants from the Federal Ministry of Education and Research (project “SARS-CoV-2Dx”), [grant number 13N15745] and the Thüringer Aufbaubank (project “Pandemie Analyse mittels Advanced Analytics Methoden”), [grant number 2021 VF 0035]. We acknowledge support by the German Research Foundation Projekt-Nr. 512648189 and the Open Access Publication Fund of the Thueringer Universitaets-und Landesbibliothek Jena.

Conflicts of interest

The authors have declared that no competing interests exist.

Data sharing statement

All genomic data (genomes and respective metadata) are available in the provided microreact project (project file available under <https://osf.io/n5qj6/> ; <https://microreact.org/project/ftR2GfjF6iXtSwbmN4ARTx-thuringianalpha-linclusters#76ir-complete-overview> ). The mobile service data used in this study can only be published in processed form (available under <https://osf.io/n5qj6/> ). The original mobile service data can not be made public due to legal reasons/ownership.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used Grammarly in order to correct general English and improve readability. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

References

- 1 Zhu N, Zhang D, Wang W, et al. (2020) **A Novel Coronavirus from Patients with Pneumonia in China, 2019** *N Engl J Med* **382**:727–33
- 2 COVID-19 Genomics UK (COG-UK) (2020) **An integrated national scale SARS-CoV-2 genomic surveillance network** *Lancet Microbe* **1**:e99–100
- 3 German ‘Corona-Surveillanceverordnung’ as issued on the 18th of January 2021 and updated on the 27th of June 2022 **German ‘Corona-Surveillanceverordnung’ as issued on the 18th of January 2021 and updated on the 27th of June 2022.**
https://www.bundesgesundheitsministerium.de/fileadmin/Dateien/3_Downloads/C/Coronavirus/
<https://www.bundesanzeiger.de/pub/publication/hEG1y5vUCsSa43vJZSP/content/hEG1y5vUCsSa43vJZSP>
 inline.
- 4 Meredith LW, Hamilton WL, Warne B, et al. (2020) **Rapid implementation of SARS-CoV-2 sequencing to investigate cases of health-care associated COVID-19: a prospective genomic surveillance study** *Lancet Infect Dis* **20**:1263–71
- 5 Page AJ, Mather AE, Le-Viet T, et al. (2021) **Large-scale sequencing of SARS-CoV-2 genomes from one region allows detailed epidemiology and enables local outbreak management** *Microb Genomics* **7** <https://doi.org/10.1099/mgen.0.000589>
- 6 Lambrou AS, Shirk P, Steele MK, et al. (2022) **Genomic Surveillance for SARS-CoV-2 Variants: Predominance of the Delta (B.1.617.2) and Omicron (B.1.1.529) Variants — United States, June 2021–January 2022** *MMWR Morb Mortal Wkly Rep* **71**:206–11
- 7 Brandt C, Krautwurst S, Spott R, et al. (2021) **poreCov-An Easy to Use, Fast, and Robust Workflow for SARS-CoV-2 Genome Reconstruction via Nanopore Sequencing** *Front Genet* **12**
- 8 Lataretu M, Drechsel O, Kmiecinski R, Trappe K, Hölzer M, Fuchs S. (2023) **Lessons learned: overcoming common challenges in reconstructing the SARS-CoV-2 genome from short-read sequencing data via CoVpipe2** *F1000Research* **12**
- 9 Rambaut A, Holmes EC, O’Toole Á, et al. (2020) **A dynamic nomenclature proposal for SARS-CoV-2 lineages to assist genomic epidemiology** *Nat Microbiol* **5**:1403–7
- 10 PANGO SARS-CoV-2 lineage overview **PANGO SARS-CoV-2 lineage overview; accessed 18th August 2023.** https://github.com/cov-lineages/lineages-website/blob/master/_data/lineage_data.full.json.
- 11 Washington NL, Gangavarapu K, Zeller M, et al. (2021) **Emergence and rapid transmission of SARS-CoV-2 B.1.1.7 in the United States** *Cell* **184**:2587–2594
- 12 Walker AS, Vihta K-D, Gethings O, et al. (2021) **Tracking the Emergence of SARS-CoV-2 Alpha Variant in the United Kingdom** *N Engl J Med* **385**:2582–5
- 13 Michaelsen TY, Bennedbæk M, Christiansen LE, et al. (2022) **Introduction and transmission of SARS-CoV-2 lineage B.1.1.7, Alpha variant, in Denmark** *Genome Med* **14**

- 14 Alpert T, Brito AF, Lasek-Nesselquist E, et al. (2021) **Early introductions and transmission of SARS-CoV-2 variant B.1.1.7 in the United States** *Cell* **184**:2595–2604
- 15 O'Toole Á, Hill V, Pybus OG, et al. (2021) **Tracking the international spread of SARS-CoV-2 lineages B.1.1.7 and B.1.351/501Y-V2 with grinch** *Wellcome Open Res* **6**
- 16 Li J, Ahmed W, Metcalfe S, et al. (2022) **Monitoring of SARS-CoV-2 in sewersheds with low COVID-19 cases using a passive sampling technique** *Water Res* **218**
- 17 Kraemer MUG, Hill V, Ruis C, et al. (2021) **Spatiotemporal invasion dynamics of SARS-CoV-2 lineage B.1.1.7 emergence** *Science* **373**:889–95
- 18 Chang S, Pierson E, Koh PW, et al. (2021) **Mobility network models of COVID-19 explain inequities and inform reopening** *Nature* **589**:82–7
- 19 Schlosser F, Maier BF, Jack O, Hinrichs D, Zachariae A, Brockmann D. (2020) **COVID-19 lockdown induces disease-mitigating structural changes in mobility networks** *Proc Natl Acad Sci* **117**:32883–90
- 20 Meintrup D, Nowak-Machen M, Borgmann S. (2022) **A Comparison of Germany and the United Kingdom Indicates That More SARS-CoV-2 Circulation and Less Restrictions in the Warm Season Might Reduce Overall COVID-19 Burden** *Life* **12**
- 21 Oh DY, Hölzer M, Paraskevopoulou S, et al. (2022) **Advancing Precision Vaccinology by Molecular and Genomic Surveillance of Severe Acute Respiratory Syndrome Coronavirus 2 in Germany, 2021** *Clin Infect Dis* **75**:S110–20
- 22 Pletz MW, Steiner A, Kesselmeier M, et al. (2023) **Introduction of mandatory masking in health care and community: experience from Jena, Germany** *Infection* <https://doi.org/10.1007/s15010-023-02015-w>
- 23 Aksamentov I, Roemer C, Hodcroft E, Neher R. (2021) **Nextclade: clade assignment, mutation calling and quality control for viral genomes** *J Open Source Softw* **6**
- 24 Ozono S, Zhang Y, Ode H, et al. (2021) **SARS-CoV-2 D614G spike mutation increases entry efficiency with enhanced ACE2-binding affinity** *Nat Commun* **12**
- 25 Du Y, Shi R, Zhang Y, et al. (2021) **A broadly neutralizing humanized ACE2-targeting antibody against SARS-CoV-2 variants** *Nat Commun* **12**
- 26 Liang R, Ye Z-W, Ong CP, et al. (2022) **The spike receptor-binding motif G496S substitution determines the replication fitness of SARS-CoV-2 Omicron sublineage** *Emerg Microbes Infect* **11**:2093–101
- 27 Kimura I, Yamasoba D, Nasser H, et al. (2022) **The SARS-CoV-2 spike S375F mutation characterizes the Omicron BA.1 variant** *iScience* **25**
- 28 Asif A, Ilyas I, Abdullah M, Sarfraz S, Mustafa M, Mahmood A. (2022) **The Comparison of Mutational Progression in SARS-CoV-2: A Short Updated Overview** *J Mol Pathol* **3**:201–18
- 29 Donzelli S, Spinella F, di Domenico EG, et al. (2022) **Evidence of a SARS-CoV-2 double Spike mutation D614G/S939F potentially affecting immune response of infected subjects** *Comput Struct Biotechnol J* **20**:733–44

- 30 Freed N, Silander O. (2021) **Freed N, Silander O. SARS-CoV2 genome sequencing protocol (1200bp amplicon “midnight” primer set, using Nanopore Rapid kit) v5. 2021** DOI:10.17504/protocols.io.btsrnn6. <https://doi.org/10.17504/protocols.io.btsrnn6>
- 31 Jundzill M, Spott R, Lohde M, Hölzer M, Viehweger A, Brandt C. (2023) **Managing and monitoring a pandemic: showcasing a practical approach for the genomic surveillance of SARS-CoV-2** *Database* 2023
- 32 Koch-Institut R. (2022) **Koch-Institut R. SARS-CoV-2 Sequenzdaten aus Deutschland. 2022; published online Oct 16.** DOI:10.5281/ZENODO.7212725. <https://doi.org/10.5281/ZENODO.7212725>

Article and author information

Riccardo Spott

Institute for Infectious Diseases and Infection Control, Jena University Hospital, Jena, Germany

For correspondence: riccardo.spott@gmail.com

ORCID ID: [0000-0002-2103-167X](https://orcid.org/0000-0002-2103-167X)

Mathias W. Pletz

Institute for Infectious Diseases and Infection Control, Jena University Hospital, Jena, Germany, Center for Sepsis Control and Care, Jena University Hospital/Friedrich Schiller University Jena, Jena, Germany

Carolin Fleischmann-Struzek

Institute for Infectious Diseases and Infection Control, Jena University Hospital, Jena, Germany, Center for Sepsis Control and Care, Jena University Hospital/Friedrich Schiller University Jena, Jena, Germany

Aurelia Kimmig

Institute for Infectious Diseases and Infection Control, Jena University Hospital, Jena, Germany

Christiane Hadlich

SMA Development GmbH - epicinsights Agentur für Künstliche Intelligenz und Big Data Analytics, Jena, Germany

Mathias Hauert

SMA Development GmbH - epicinsights Agentur für Künstliche Intelligenz und Big Data Analytics, Jena, Germany

Mara Lohde

Institute for Infectious Diseases and Infection Control, Jena University Hospital, Jena, Germany
ORCID ID: [0009-0009-1718-7231](https://orcid.org/0009-0009-1718-7231)

Mateusz Jundzill

Institute for Infectious Diseases and Infection Control, Jena University Hospital, Jena, Germany

Mike Marquet

Institute for Infectious Diseases and Infection Control, Jena University Hospital, Jena, Germany
ORCID ID: [0000-0003-4344-8289](https://orcid.org/0000-0003-4344-8289)

Petra Dickmann

Department of Anaesthesiology and Intensive Care, Jena University Hospital, Jena, Germany

Ruben Schüchner

Thuringian State Authority for Consumer Protection, Department Health Protection

Martin Hölzer

Methodology and Research Infrastructure, Genome Competence Center (MF1), Robert Koch Institute, Berlin, Germany

Denise Kühnert

Centre for Artificial Intelligence in Public Health Research, Robert Koch Institute, Berlin, Germany, Transmission, Infection, Diversification and Evolution Group, Max Planck Institute for Geoanthropology, 07745 Jena, Germany

Christian Brandt

Institute for Infectious Diseases and Infection Control, Jena University Hospital, Jena, Germany

Copyright

© 2024, Spott et al.

This article is distributed under the terms of the [Creative Commons Attribution License](#), which permits unrestricted use and redistribution provided that the original author and source are credited.

Editors

Reviewing Editor

Amy Wesolowski

Johns Hopkins Bloomberg School of Public Health, Baltimore, United States of America

Senior Editor

Jos van der Meer

Radboud University Medical Centre, Nijmegen, Netherlands

Reviewer #1 (Public Review):

Summary:

In "1 Exploring the Spatial Distribution of Persistent SARS-CoV-2 Mutations -Leveraging mobility data for targeted sampling" Spott et al. combine SARS-CoV-2 genomic data alongside granular mobility data to retrospectively evaluate the spread of SARS-CoV-2 alpha lineages throughout Germany and specifically Thuringia. They further prospectively identified districts with strong mobility links to the first district in which BQ.1.1 was observed to direct additional surveillance efforts to these districts. The additional surveillance effort resulted in the earlier identification of BQ.1.1 in districts with strong links to the district in which BQ.1.1 was first observed.

Strengths:

There are two important strengths of this work. The first is the scale and detail in the data that has been generated and analyzed as part of this study. Specifically, the authors use 6,500 SARS-CoV-2 sequences and district-level mobility data within Thuringia. I applaud the authors for making a subset of their analyses public e.g. on the associated micro react page.

Further, the main focus of the article is on the potential utility of mobility-directed surveillance sequences. While I may certainly be mistaken, I have not seen this proposed elsewhere, at least in the context of SARS-CoV-2. The authors were further able to test this concept in a real-world setting during the emergence of BQ.1.1. This is a unique real-world evaluation of a novel surveillance sequencing strategy and there is considerable value in publishing this analysis.

Weaknesses:

The article is quite strong and I find the analyses to generally be rigorous. However, there are places where I believe the text should be modified to slightly weaken the conclusions drawn from the presented analyses. Specific examples include:

- It seems the mobility-guided increased surveillance included only districts with significant mobility links to the origin district and did not include any "control" districts (those without strong mobility links). As such, you can only conclude that increasing sampling depth increased the rate of detection for BQ.1.1., not necessarily that doing so in a mobility-guided fashion provided an additional benefit. I absolutely understand the challenges of doing this in a real-world setting and think that the work remains valuable even with this limitation, but I would like the lack of control districts to be more explicitly discussed.

- Line 313: While this work has reliably shown that the spread of Alpha was slower in Thuringia, I don't think there have been sufficient analyses to conclude that this is due to the lack of transportation hubs. My understanding is that only mobility within Thuringia has been evaluated here and not between Thuringia and other parts of Germany.

- Line 333 (and elsewhere): I'm not convinced, based on the results presented in Figure 2, that the authors have reliably identified a sampling bias here. This is only true if you assume (as in line 235) that the variant was in these districts, but that hasn't actually been demonstrated here. While I recognize that for high-prevalence variants there is a strong correlation between inflow and variant prevalence, low-prevalence variants by definition spread less and may genuinely be missing from some districts. To support this conclusion that they identified a bias, I'd like to see some type of statistical model that is based e.g. on the number of sequences, prevalence of a given variant in other districts, etc. Alternatively, the language can be softened ("putative sampling bias").

<https://doi.org/10.7554/eLife.94045.1.sa1>

Reviewer #2 (Public Review):

In the manuscript, the authors combine SARS-CoV-2 sequence data from a state in Germany and mobility data to help in understanding the movement of the virus and the potential to help decide where to focus sequencing. The global expansion in sequencing capability is a key outcome of the public health response. However, there remains uncertainty about how to maximise the insights the sequence data can give. Improved ability to predict the movement of emergent variants would be a useful public health outcome. Also knowing where to focus sequencing to maximising insights is also key. The presented case study from one State in Germany is therefore a useful addition to the literature. Nevertheless, I have a few comments.

One of the key goals of the paper is to explore whether mobile phone data can help predict the spread of lineages. However, it appears unclear whether this was actually addressed in the analyses. To do this, the authors could hold out data from a period of time, and see whether they can predict where the variants end up being found.

The abstract presents the mobility-guided sampling as a success, however, the results provide a much more mixed result. Ultimately, it's unclear what having this strategy really achieved. In a quickly moving pandemic, it is unclear what hunting for extra sequences of a specific, already identified, variant really does. I'm not sure what public health action would result, especially given the variant has already been identified.

Relatedly, it is unclear to me whether simply relying on spatial distance would not be an alternative simpler approach than mobile phone data. From Figure 2, it seems clear that a simple proximity matrix would work well at reconstructing viral flow. The authors could compare the correlation of spatial, spatial proximity, and CDR data.

<https://doi.org/10.7554/eLife.94045.1.sa0>